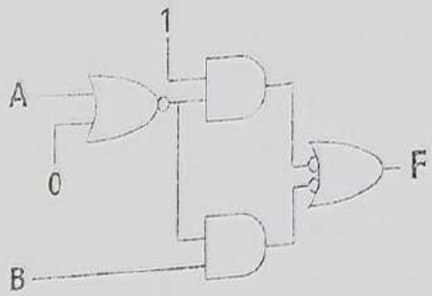
...m # 1: Answer the following Multiple-choice questions: [10 x 1pt]

The output **F** of the combinational circuit shown to the right is:

A. $\overline{A} + B$ B. 1 C. $A + \overline{B}$ D. $\overline{A} + \overline{B}$



2- The range of signed decimal numbers that can be represented by 9 bits is:

A. −256 to +255 B. −128 to +128 C. −64 to +63 D. None of the above

3- The Hexadecimal representation of $(45.0625)_{10}$ is:

A. $(2C.4)_{16}$ B. $(2D.02)_{16}$ C. $(2E.04)_{16}$ D. None of the above

4- In 16-bit 2's complement representation, the decimal number -34 is:

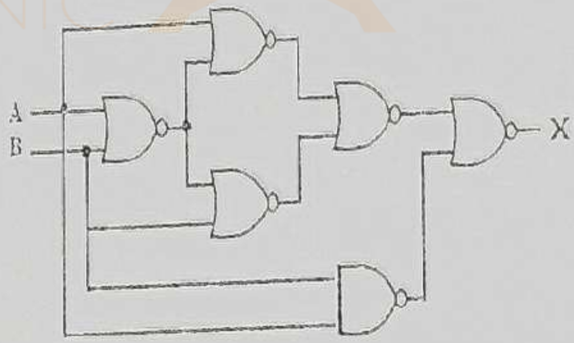
A. 0000000000100010
B. 1111111111011101
C. 1111111111011110
D. None of the above

5- The result of $(F0ACD472)_{16}$ + $(89D34FA9)_{16}$ is:

A. $(17A80231B)_{16}$
B. $(17B80241B)_{16}$
C. $(17A80251B)_{16}$
D. None of the above

6- The combinational circuit shown in the right is:

A. 1
B. $\overline{A}$
C. 0
D. $\overline{B}$



7- The equivalent SOP canonical form for the following logical expression

$$F(A,B,C) = AB + C \text{ is:}$$

A) $F(A,B,C) = ABC + \bar{A}BC + A\bar{B}C + \bar{A}\bar{B}C$

B) $F(A,B,C) = ABC + A\bar{B}C + \bar{A}\bar{B}C + AB\bar{C}$

C) $F(A,B,C) = ABC + \bar{A}BC + A\bar{B}C + \bar{A}\bar{B}C + AB\bar{C}$

D) None of the above

ing **7-bit signed** numbers (two's complement), which of the following decimal additions causes **overflow**?

A. 45 + 13

B. -24 + 15

C. -32 + -33

D. None of the above

9- Simplified form of the Boolean expression (X + Y + XY) (X + Z) is:

A) X + Y + Z B) XY + YZ C) X + YZ D) None of the above

10- If $F(x, y, z, w) = \pi\,(0, 3, 5, 7, 8, 9, 10, 15)$, the complement of F ($\overline{F}$) is:

A. $\pi\,(1, 2, 4, 6, 11, 12, 13, 14)$

B. $\pi\,(1, 2, 4, 6, 11, 12, 10, 14)$

C. $\sum(0, 3, 5, 7, 8, 9, 10, 15)$

D. None of the above

*** Write the correct answers to problem # 1 in the table below:

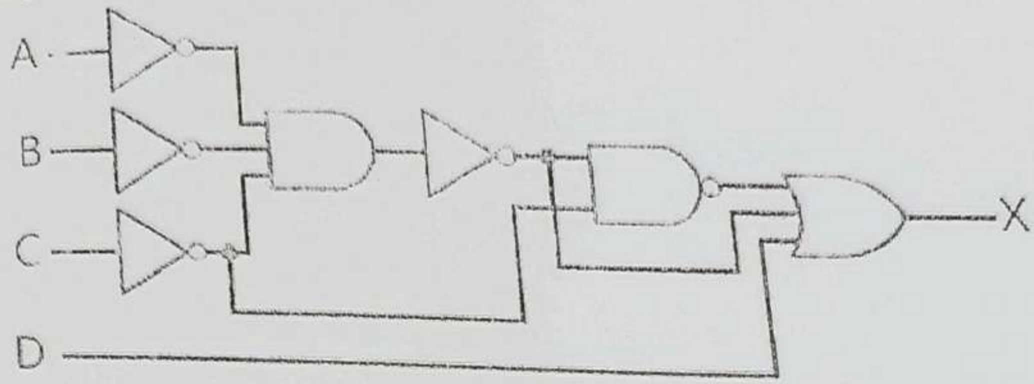
<u>Problem#2:</u> Find all possible **<u>most simplified SOP (Sum of Products)</u>** expressions for

$$F(A.B.C.D) = \sum(0, 2, 4, 5, 6, 8, 10, 15)$$

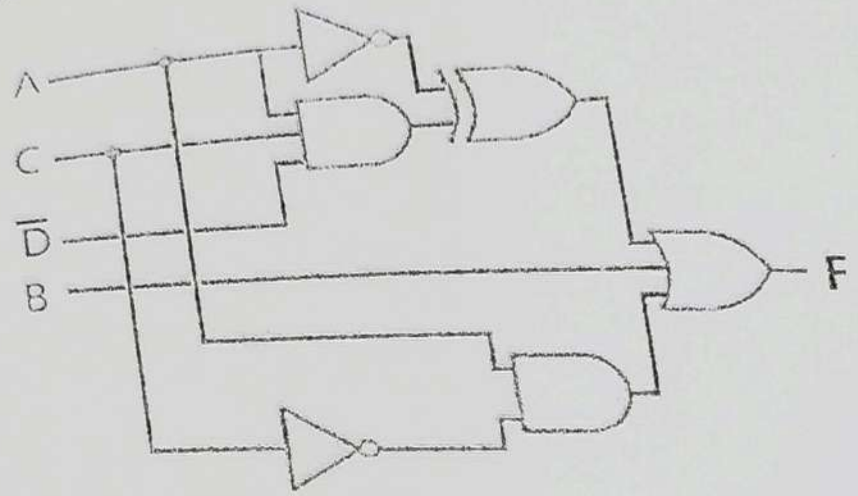
together with the don't care conditions $d(A,B,C,D) = \sum(7, 13, 14)$

You must <u>show all steps,</u> including the drawing of the cells and grouping cells together, etc. [8pts]

...roblem # 3: Based on the below combinational circuit, find the minimal form of **X** function using Boolean algebra rules. [6pts]



Problem # 4: Implement the combinational circuit below using only **NAND** gates. [6pts]



Good Luck